

Omer Pinto

☎ (+972) 54-534-7868 | ✉ omerp2006@gmail.com | 🌐 Omer-Pinto | 📄 omer-pinto-a940a8122

Summary & Current Focus

- Senior Software Engineer with **15+ years** building **backend systems, database internals, & distributed platforms**.
- Currently developing **Python** microservices and orchestration systems at BigArch while actively expanding into **AI Engineering**.
- Develop **agentic workflows** using **LangGraph / LangChain** through personal projects & professional AI engineering coursework.
- Strong foundation in **algorithms, performance tuning, & systems architecture** across C++, Python, Java, & cloud platforms.
- Seeking **AI Engineering roles** focused on intelligent products and **LLM-powered applications**.

Experience

BigArch

SENIOR SW ENGINEER

2024 - Present

- Backend development of a large-scale **graph orchestration platform** for HPC with **Python, MongoDB, RabbitMQ & microservices**.
- Architect and deploy containerized services in private-cloud environment using **Kubernetes** and **Docker**.
- Achieved **60% performance gains** through system-wide optimization, transforming strained system into stable, scalable platform.
- Lead system-wide architectural refactors executed with **zero downtime** and full service continuity.
- **AI Initiatives:** Evaluated multiple LLMs in air-gapped system. Built custom shim server enabling LLM tool integration.
- Drive AI-assisted development adoption through practical MVPs & team enablement (More on AI work & self-study in Projects section)

Microsoft

SENIOR SW ENGINEER

2023 - 2024

- Engineer in the **vulnerability management** group, developing components in the endpoint security pipeline.
- Built **C# microservices** and data pipelines using Azure services (Blobs, Queues, CosmosDB, KustoDB).

Yahoo

PRINCIPAL SW ENGINEER

2022 - 2023

- Contributor to the **native ads** platform, developing **Java** services and **Hadoop/Spark** ETL pipelines.
- Partnered with research team to implement **ML/DL-driven ad-scoring algorithms** for multi-variable ranking system.
- Managed codebase migration and team handoff following **project cancellation**.

qSpark

SENIOR SW ENGINEER

2021 - 2022

- Delivered ultra-low-latency trading infrastructure. Implemented exchange protocols (execution & market data) using modern C++ with template meta-programming.
- Designed scalable subsystem enabling 3rd-party trading algorithms integration.
- Handled production issues with global exchanges and conducted market-structure research (SEC/FINRA rules).

Toga Networks

HANDS-ON ARCHITECT

2019 - 2021

- Worked across multiple projects in the **cloud and IoT R&D** group, driving architecture and implementation.
- Performed **market and technology research** and delivered high-level designs for innovative systems.
- Led research & design of semantic analytics engine for IoT data, including reasoning capabilities across data model and query layers.
- Developed a log-analysis algorithm using **exponential backoff**, improving system performance.

SiSense

SENIOR ENGINEER, TECH LEAD & ARCHITECT, CTO OFFICE

2014 - 2018

- **Hand-picked by CTO** for his own private team researching next-gen analytics & developing features on open-source DB project.
- Architected and built cross-platform **ETL query engine** from scratch as sole developer; recruited and led team as tech lead.
- Engine optimized for low RAM footprint, 100% multicore CPU utilization (any scale-up) and distributed execution (scale-out), implemented using C++14/17 template meta-programming.
- Implemented core DB features and system optimizations using multicore algorithms and cache-aware data structures.
- Designed thread management solution improving query engine performance by order of magnitude.
- Optimized query operations (join, group) on multi-billion record datasets
- Implemented statistical algorithms from academic papers.
- Contributed by implementing innovative complex DB compiler internals and execution engine features.

Early years - Allot Communications, ZRRO Technologies

FIRMWARE / EMBEDDED ENGINEER

2011-2014

- Design & implementation of **real-time algorithms & data structures** in a resource-restricted, **performance-critical** environment.
- Network protocols, coding of algorithms & systems in **C & C++** using extreme **memory/cpu optimizations** on **Linux** os.

Technical Skills

- **Python & Backend Development:** Production experience with microservices architecture, distributed systems (MongoDB, RabbitMQ, Kafka). RESTful APIs, async programming, performance optimization. Docker, Kubernetes for containerized deployments.
- **AI & LLM Systems:** Hands-on development with agentic frameworks (LangGraph, LangChain). Prompt engineering, multi-agent workflows, MCP protocol integration. Experience deploying LLM infrastructure in air-gapped environments. Ongoing coursework and hands-on project development.
- **Database Systems & Internals:** Deep experience implementing DB internals (query parsers, execution engines, optimizers, caching layers). Designed and built ETL engine from scratch. Working knowledge of PostgreSQL, MongoDB, ClickHouse, Druid, KustoDB. Query language DSL design.
- **Additional Languages:** Java (Spring, Maven, Hadoop/Spark), C# (Azure microservices), C++ (template metaprogramming, low-latency systems, multi-threading). Bash scripting.
- **Cloud & Distributed Systems:** Azure services (Blobs, Queues, CosmosDB, KustoDB), AWS (EC2, S3). Hadoop/Spark pipelines, distributed orchestration platforms.
- **Development Practices:** Git, CI/CD (Jenkins, ADO), Agile/Scrum, unit testing, system design, performance profiling.

Education

The Open University of Israel

Ra'anana, Israel

B.Sc IN MATHEMATICS AND COMPUTER SCIENCE

2007 - 2010

- Graduated **summa cum laude** with GPA of **97/100**.
- Awarded place on University President's Excellence List (annual GPA ≥ 95) for four consecutive years.
- Achievement recognized in university educational portal [article](#).

Independent Work & Technical Writing

AI Projects, Udemy AI Learning & Technical Writing

- **LangGraphWrappers:** Infrastructure layer for building LangGraph agents with reusable components (nodes, edges, tools, structured outputs). Demonstrates systems-level thinking applied to agent development frameworks. [GitHub](#).
- **StocksMarketRecommender:** Multi-agent workflow with Yahoo Finance MCP integration. Orchestrates multiple LangGraph flows using LangGraphWrappers library. [GitHub](#).
- **SoccerSmartBet (ongoing):** Real-time betting analysis system comparing AI vs. human strategies. Demonstrates high level architecture of complex independent agentic workflows using LangGraph infrastructure. [GitHub](#).
- Completed **AI Engineer Agentic Track: The Complete Agent & MCP Course** (Udemy, 2025) - comprehensive training in AI agents, multi-agent systems, MCP protocol, prompt engineering, and AI systems design.
- Published technical series on database implementation and C++ techniques on LinkedIn, republished by [EuropeClouds](#) (2020-2021). Created [C++ TMP Tutorial](#) referenced in articles.